A 1D Shallow-Water Test Problem: Tidal Forcing, Wetting–Drying, and an Internal Berm

Abstract

We describe a small 1D implementation of the nonlinear shallow-water equations intended as a test bed for the boundary treatments discussed in the companion note on tide–storm–surge modeling. The domain is a sloped beach with a reflecting landward wall, forced on the seaward side by a prescribed sinusoidal free-surface elevation. The model includes wetting and drying of the intertidal zone and, optionally, an internal rectangular berm across which water spills. The spatial discretization is a first-order finite-volume scheme using Audusse's hydrostatic reconstruction together with a Rusanov (local Lax–Friedrichs) numerical flux; time integration uses a strong-stability preserving Runge–Kutta scheme (SSP-RK2 / Heun). Mass balance, free-surface extrema, and shoreline excursion are monitored. Representative simulations are presented both without and with the internal berm.

1 Problem statement

Let $x \in [0, L]$ be the horizontal coordinate, $z_b(x)$ the bed elevation (positive upward), and $\eta(x,t)$ the free-surface elevation so that the total water depth is

$$H(x,t) = \eta(x,t) - z_b(x), \qquad H \geq 0. \tag{1}$$

With u the depth-averaged velocity and $q = Hu$ the discharge, the one-dimensional shallow-water equations are

$$\begin{aligned} \partial_t H + \partial_x (Hu) &= 0, \\ \partial_t (Hu) + \partial_x \Big(Hu^2 + \tfrac{1}{2} g H^2 \Big) &= -gH\, \partial_x z_b. \end{aligned} \tag{2}$$

We consider the geometry sketched in Figure 1. The bed decreases linearly from a land-side elevation z_{beach} down to a seaward depth z_{deep}. The left boundary at $x = 0$ is a reflecting wall representing the land; the right boundary at $x = L$ is an open-ocean-type boundary where the free surface is prescribed as a pure tide:

$$\eta(L,t) = A \sin(\omega t), \qquad \omega = \frac{2\pi}{T}. \tag{3}$$

Representative physical parameters used in the simulations below are collected in Table 1.

2 Finite-volume discretization

The domain is partitioned into N cells of width $\Delta x = L/N$ with centers $x_j = (j - \frac{1}{2})\Delta x$, $j = 1, \ldots, N$. The discrete state is the pair of cell averages $(H_j, q_j) = (H_j, (Hu)_j)$. Fluxes are defined at the $N{+}1$ interfaces $x_{j+1/2} = j\Delta x$.

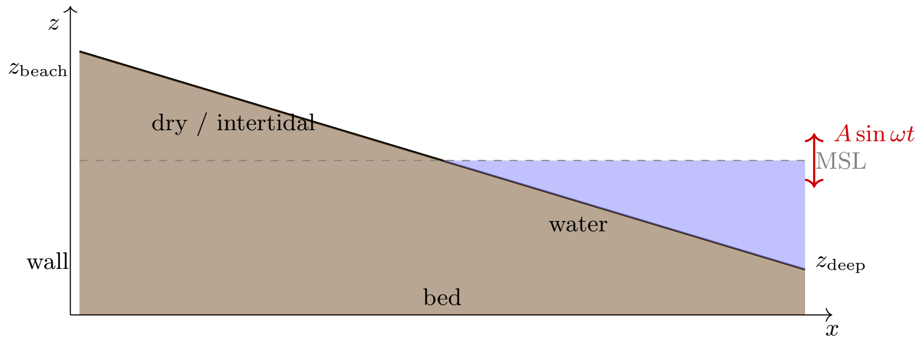

Figure 1: Schematic of the 1D test problem: sloping bed, reflecting wall at $x = 0$, tidal forcing at $x = L$. The intertidal portion of the bed wets and dries as the tide rises and falls.

Table 1: Baseline parameters.

Symbol	Value	Description
L	10 000 m	domain length
z_{beach}	+3 m	bed at $x = 0$
z_{deep}	−7 m	bed at $x = L$
A	0.4 m	tidal amplitude
T	1800 s	tidal period
g	9.81 m s^{-2}	gravitational acceleration
N	400	cells
Δx	25 m	cell size
CFL	0.4	target Courant number
H_{tol}	10^{-4} m	wet/dry threshold

For any conservative SWE flux $\mathbf{F}(U) = (Hu,\ Hu^2 + \frac{1}{2}gH^2)$, we use the local Lax–Friedrichs (Rusanov) numerical flux

$$\mathbf{F}^{\mathrm{R}}(U_L, U_R) = \tfrac{1}{2}\big(\mathbf{F}(U_L) + \mathbf{F}(U_R)\big) - \tfrac{1}{2}\alpha\,(U_R - U_L), \tag{4}$$

where the wave-speed estimate is $\alpha = \max\big(|u_L| + \sqrt{gH_L},\ |u_R| + \sqrt{gH_R}\big)$.

2.1 Audusse hydrostatic reconstruction

To preserve the lake-at-rest equilibrium and to remain well-behaved at wet/dry interfaces across bed steps, we apply the hydrostatic reconstruction of Audusse et al. (2004). At each interior face $j+\frac{1}{2}$, let $z^*_{j+1/2} = \max(z_{b,j},\ z_{b,j+1})$, and define the reconstructed left- and right-face depths

$$H^{*,-}_{j+1/2} = \max\big(0,\ \eta_j - z^*_{j+1/2}\big), \qquad H^{*,+}_{j+1/2} = \max\big(0,\ \eta_{j+1} - z^*_{j+1/2}\big). \tag{5}$$

Velocities are carried unchanged from the cell states, so the reconstructed interface discharges are $(Hu)^{*,\mp}_{j+1/2} = H^{*,\mp}_{j+1/2}\,u_{j\,\mathrm{or}\,j+1}$.

The numerical flux is evaluated on the reconstructed states, $\mathbf{F}^*_{j+1/2} = \mathbf{F}^{\mathrm{R}}\big(U^{*,-}_{j+1/2}, U^{*,+}_{j+1/2}\big)$, and the bed step is accounted for by an asymmetric correction to the momentum flux: the value seen by the left cell and by the right cell of the face are, respectively,

$$F^L_{Hu,\,j+1/2} = F^*_{Hu,\,j+1/2} + \tfrac{1}{2}g\big(H_j^2 - (H^{*,-}_{j+1/2})^2\big), \tag{6}$$
$$F^R_{Hu,\,j+1/2} = F^*_{Hu,\,j+1/2} + \tfrac{1}{2}g\big(H_{j+1}^2 - (H^{*,+}_{j+1/2})^2\big). \tag{7}$$

The mass flux is the same on both sides. With this construction, no explicit bed-slope source term is required: in particular, when η is constant, $H^{*,-} = H^{*,+}$ at every face, the Rusanov flux gives pure hydrostatic pressure, and the two corrections cancel cell-by-cell to machine precision regardless of z_b.

2.2 Semi-discrete update

Let $\mathcal{L}(U)_j$ denote the spatial discretization, written component-wise as

$$\mathcal{L}(U)_j = \begin{pmatrix} -\dfrac{1}{\Delta x}\big(F^*_{H,\,j+1/2} - F^*_{H,\,j-1/2}\big) \\ -\dfrac{1}{\Delta x}\big(F^L_{Hu,\,j+1/2} - F^R_{Hu,\,j-1/2}\big) \end{pmatrix}. \tag{8}$$

2.3 Time integration

We use the two-stage strong-stability-preserving Runge–Kutta scheme (Heun):

$$\begin{aligned} U^{(1)} &= U^n + \Delta t\,\mathcal{L}(U^n, t^n), \\ U^{n+1} &= \tfrac{1}{2}U^n + \tfrac{1}{2}\Big[U^{(1)} + \Delta t\,\mathcal{L}(U^{(1)}, t^n + \Delta t)\Big]. \end{aligned} \tag{9}$$

The time step is set by the CFL condition $\Delta t = \mathrm{CFL}\,\dfrac{\Delta x}{\max_j(|u_j| + \sqrt{gH_j})}$.

2.4 Wet/dry floor

After each stage, we enforce $H \geq H_{\text{tol}}$ by clipping: any cell with $H < H_{\text{tol}}$ is reset to H_{tol} and its discharge zeroed. This is the only non-conservative operation in the scheme and is the dominant source of mass drift discussed below.

3 Boundary conditions

Left wall. A no-flow wall is enforced using a mirror state: $(H_{\text{ghost}}, (Hu)_{\text{ghost}}) = (H_1, -q_1)$ with $z_{b,\text{ghost}} = z_{b,1}$. The mass flux is identically zero at the wall face; the momentum flux reduces to pure hydrostatic pressure at rest.

Right open-ocean boundary. A ghost cell is constructed with bed $z_{b,\text{ghost}} = z_{b,N}$, free-surface elevation prescribed by (3), and velocity extrapolated from the interior (Neumann). The flux at face $N+\frac{1}{2}$ is the Rusanov flux between the interior state and this ghost state.

4 Diagnostics

Let $\mathcal{M}(t) = \sum_j H_j(t)\,\Delta x$ denote the total mass of water in the domain. For a domain with a reflecting left wall, exact discrete conservation implies

$$\mathcal{M}(t) - \mathcal{M}(0) + \int_0^t F^*_{H,\,N+1/2}(\tau)\,d\tau \overset{!}{=} 0. \tag{10}$$

Any departure from this identity is the *mass residual* and is caused exclusively by the wet/dry clipping. We track $\mathcal{M}(t)$, the integrated boundary flux, and their sum (the residual). In addition we track the maximum and minimum free-surface elevation and the leftmost wet cell (shoreline position). With an internal structure present, we also record the mass flux through its crest.

5 Results without internal obstacle

In this first configuration the bed is a plain linear slope, with no internal structure. Figure 2 shows the final state of the simulation after eight tidal periods; the upper panel displays the full domain and the lower panel a zoom on the moving shoreline region. Figure 3 shows the time histories of the mass balance, the mass residual, the surface extrema, and the shoreline position.

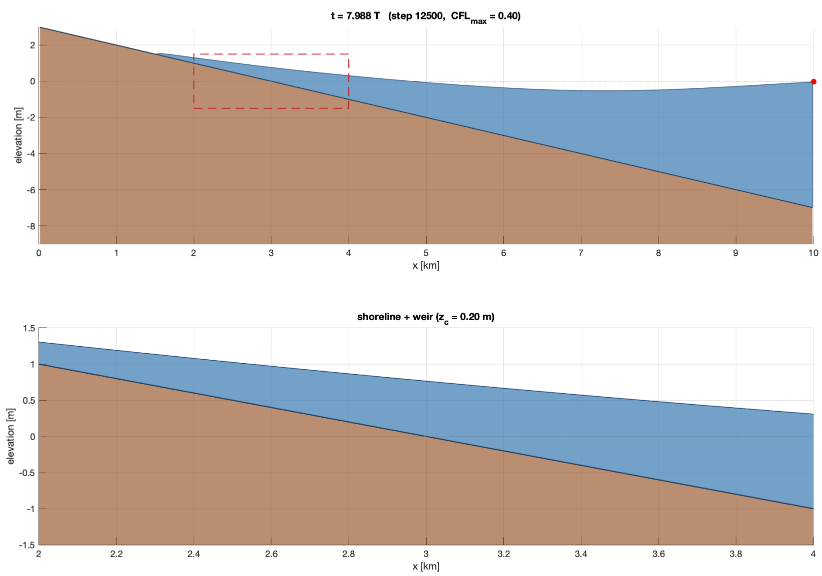

Figure 2: Final state, configuration without internal obstacle. Top: full domain. Bottom: zoom on the moving shoreline. The dashed line marks mean sea level.

The shoreline moves by several hundred meters over the tidal cycle, the maximum and minimum elevations track the forcing with negligible phase lag given the short domain, and the mass residual accumulates slowly and monotonically — consistent with the one-sided nature of the wet/dry clipping.

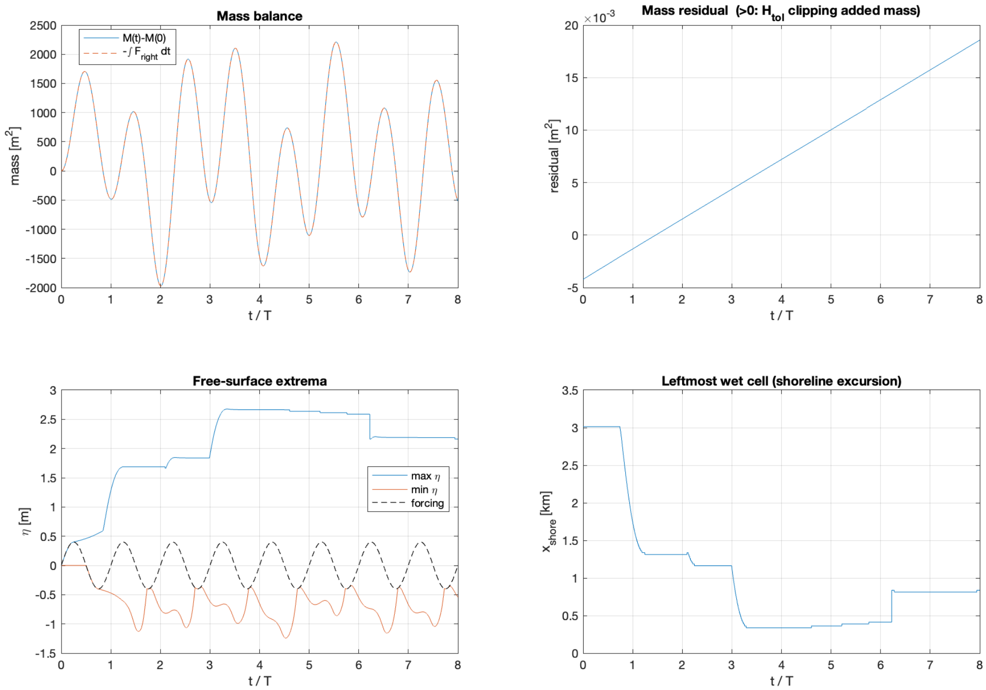

Figure 3: Diagnostics, configuration without internal obstacle. Clockwise from top-left: mass balance check against the integrated right-boundary flux; mass residual (growth is due to the wet/dry floor); surface extrema compared with the forcing; shoreline excursion.

6 Results with an internal berm

A rectangular berm is inserted as a local modification of the bed:

$$z_b(x) \leftarrow z_c \quad \text{for } |x - x_w| \leq w_h, \tag{11}$$

where x_w is the berm center, w_h is its half-width, and z_c is the crest elevation. The berm thus presents two vertical bed steps of size $z_c - z_b^{\text{nat}}(x_w \pm w_h)$. The Audusse reconstruction handles these steps without generating spurious oscillations in the lake-at-rest state, and with only small transients during active overtopping.

For the simulations reported here, $x_w = 4000$ m, $w_h = 75$ m, and $z_c = 0.2$ m. The berm is thus about 1.2 m tall above the surrounding natural bed and 150 m wide. With the tidal amplitude $A = 0.4$ m, the seaward η oscillates through ± 0.4 m so the crest is overtopped only during the upper 0.2 m of each high tide.

Figure 4 shows a snapshot of the flow with the berm in place. A trapped pool is visible landward of the berm between the natural shoreline and the structure; this pool is filled in steps by overtopping events at successive high tides. Figure 5 shows the corresponding diagnostics; the bottom-right panel overlays the mass flux through the berm crest onto the shoreline excursion, making the discrete overtopping events visible.

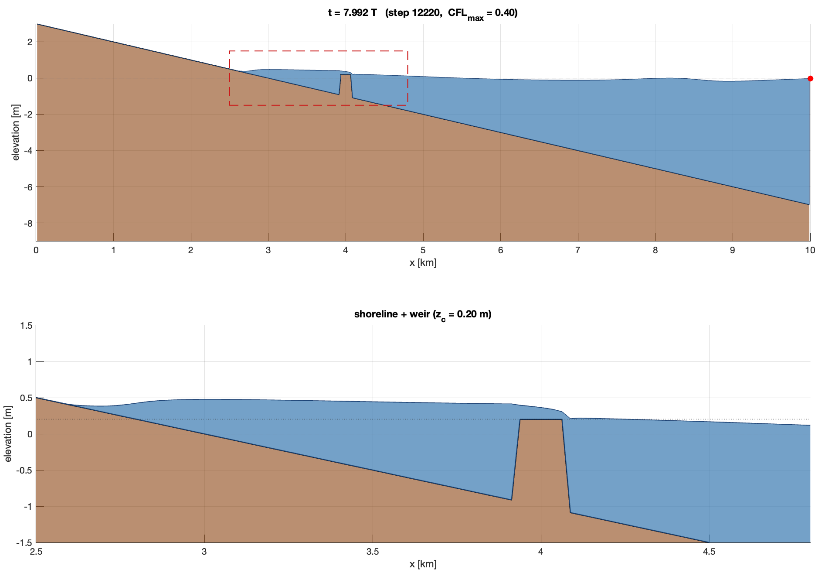

Figure 4: Final state, configuration with a rectangular berm at $x_w = 4$ km. Top: full domain — the berm (crest at $z_c = 0.2$ m) is visible as the small rectangular rise near $x = 4$ km. Bottom: zoom over $x \in [2.5, 4.8]$ km. A dotted horizontal line marks the crest elevation.

7 Discussion and limitations

The combination of Audusse reconstruction, Rusanov flux, and SSP-RK2 gives a robust, positivity-preserving, well-balanced scheme with minimal implementation complexity. The main limitations of the present implementation are:

- First-order spatial accuracy. Extension to a second-order scheme would require a piecewise-linear reconstruction of η and q inside each cell together with a slope limiter.
- A hard wet/dry floor H_{tol} that, although small, generates a monotonic and unbounded mass residual. Switching to the exponential parameterization $H(\phi) = H_{\min} + e^{\phi}$ of the companion note would eliminate this artefact entirely.
- The internal structure is resolved in the bed rather than through a classical overtopping formula. This is convenient but couples the "structure" to the grid resolution; a sub-grid weir would be modeled via an internal flux condition such as $q_n = C_w \max(0, \eta_1 - \eta_2 - z_c)^{3/2}$.
- The reflecting left wall suppresses any river-side inflow. A river or discharge-specified boundary would be a straightforward extension.

Reproducing the figures

The simulation is implemented in the script `swe_1d.m`. The figure filenames referenced above are produced automatically according to the value of the flag `has_weir`: setting it to `false` writes `swe_noweir_final.png` and `swe_noweir_diag.png`; setting it to `true` writes `swe_weir_final.png` and `swe_weir_diag.png`. A movie of the run is also produced as `swe_shoreline.avi`.

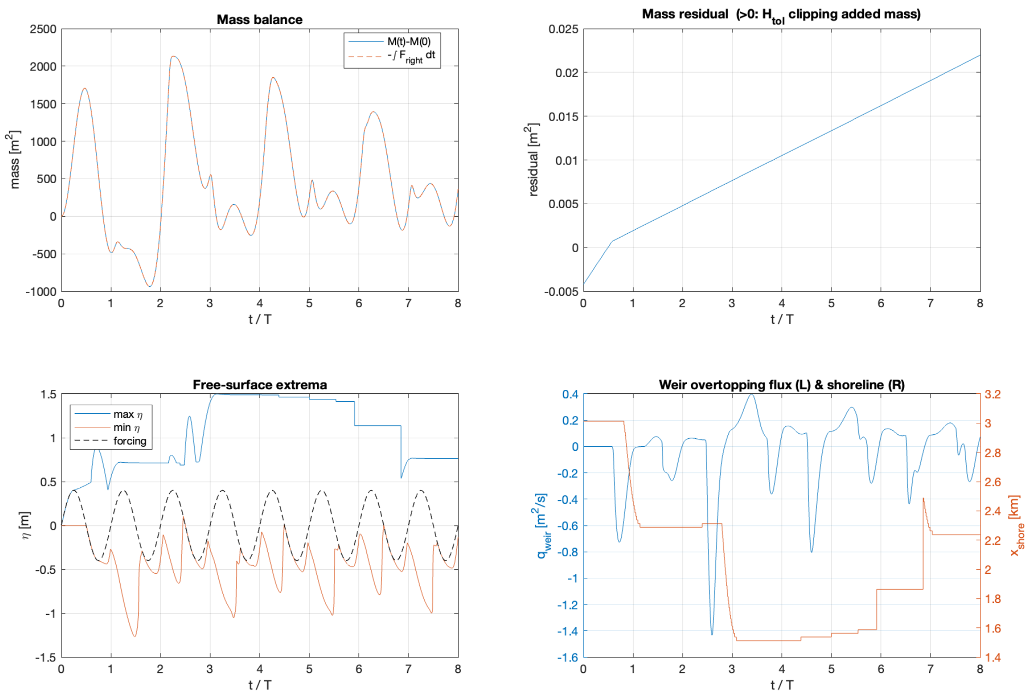

Figure 5: Diagnostics with internal berm. The bottom-right panel uses a dual-axis plot: the left axis shows the crest mass flux (spikes at each overtopping event), the right axis the shoreline position.